$$\hat{z}_t = \phi\big(x_{t-k:t-1}, a_{t-k:t-1}\big)\,,$$

$$k = 50, \qquad \hat{z}_t \in \mathbb{R}^8,$$

$$H_t = \begin{bmatrix} x_{t-50}^\top & a_{t-50}^\top \\ \vdots & \vdots \\ x_{t-1}^\top & a_{t-1}^\top \end{bmatrix} \in \mathbb{R}^{50\times 42},$$

$$\mathcal{L}_\phi = \|\hat{z}_t - \mu(e_t)\|_2^2\,.$$